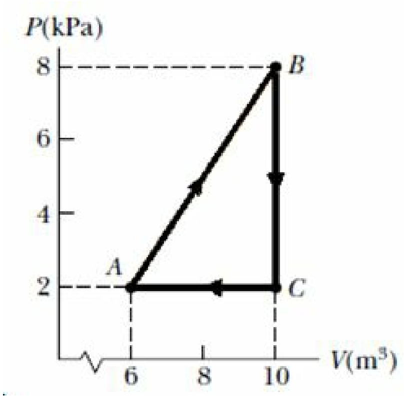
CBSE Test Paper 01

Chapter 12 Thermodynamics

1. Consider the cyclic process depicted in Figure If Q is negative for the process BC, and if ΔE_{int} is negative for the process CA, what are the signs of W and ΔE_{int} that are associated with BC? **1**



   a. 0, +
   b. 0, -
   c. +, -
   d. 0, -

2. One of the most efficient engines ever built (actual efficiency 42.0 percent) operates between 430° C and 1 870° C. How much power does the engine deliver if it absorbs1.40 $\times$ 10^5 J of energy each second from the hot reservoir? **1**

   a. 52.8 kW
   b. 62.8 kW
   c. 58.1 kW
   d. 58.8 kW

3. A power plant that would make use of the temperature gradient in the ocean has been proposed. The system is to operate between 5.00° C (water temperature at a depth of about 1 km) and 20.0° C (surface water temperature).What is the maximum efficiency of such a system? **1**

   a. 5.02 percent
   b. 5.12 percent

c. 5.45 percent
d. 5.32 percent

4. A 2.00-mol sample of helium gas initially at 300 K and 0.400 atm is compressed isothermally to 1.20 atm. Assuming the behavior of helium to be that of an ideal gas, find the energy transferred by heat **1**

   a. -5.38 kJ
   b. -5.28 kJ
   c. -5.58 kJ
   d. -5.48 kJ

5. Internal energy of a system is **1**

   a. a complex variable
   b. a random variable
   c. a state variable
   d. a discrete variable

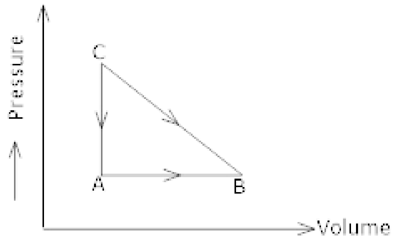
6. State Carnot's Theorem? **1**

7. State zeroth law of thermodynamics? **1**

8. Plot a graph between internal energy U and Temperature (T) of an ideal gas. **1**

9. What is thermodynamics? State its important characteristics. **2**

10. Consider the cyclic process ABCA on a sample 2 mol of an ideal gas as shown. The temperature of the gas at A and B are 300K and 500K respectively. Total of 1200 J of heat is drawn from the sample. Find the work done by the gas in part BC. **2**



11. A gas is filled in a cylinder at 300 K. Calculate the temperature upto which it should be heated so that its volume becomes $\frac{4}{3}$ of its initial volume. **2**

12. An indirectly heated filament is radiating maximum energy of wavelength 2.16×10^{-7}m. Find the net amount of heat energy lost per second per unit area, the temperature of surrounding air is 13^0C. Given b = 2.88×10^{-3} mk, $\sigma = 5.77 \times 10^{-8}$ J / sec/ m^2 / k^4 ? **3**

13. If for hydrogen C_P - C_V = a and for oxygen C_P - C_V = b where C_P & C_V refer to specific heat at constant pressure and volume then what is the relation between a and b? **3**

14. How do you derive Newton's law of cooling from Stefan's law? **3**

15. Derive the equation of state for adiabatic change. **5**